

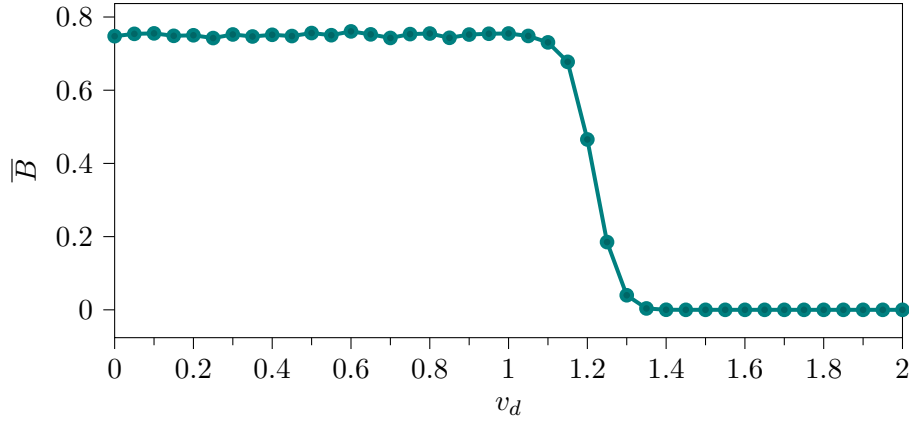


Figure 5: Backflow order parameter on variation of impurity drift velocity. Here we set $t_1 = 100$, $t_2 = 1000$ and $p = 0.1$. This is the result of an average over 10^4 disorder realizations.

the inverse swap rate $1/p$, we expect that, at t_2 , the impurity has equilibrated with the local medium and will follow the dynamics of the nearest medium qubit. We are therefore interested in the weight of the OTOC in the medium at the position of the impurity; after disorder averaging that position is $\overline{x_I} = v_d t$.

To determine the OTOC at this position, we use the approach detailed in Ref. [68]. There, the average OTOC is considered by choosing the unitary bricks from the Clifford group and using the fact that the Clifford group is a 3-design [76]. This ensures the average over the Clifford group is the same as the average over the Haar group for the OTOC. Clifford circuits map Pauli strings to Pauli strings, and since we consider an initial Pauli X operator, this operator remains a Pauli string for all time. From the definition of the OTOC we have $B(v_d, t) = 1$ if the operator $X^{(I)}(t)$ has character of X or Z on the impurity and otherwise equals zero. On disorder averaging if the operator has non-trivial (i.e. non-identity) support on a particular site then it has equal probability of having X , Y or Z character. Therefore we have that $\overline{B(v_d, t)} = \frac{2}{3} n_I(v_d, t)$, where $n_I(v_d, t)$ is a density that is equal to one if $X^{(I)}(t)$ has non-trivial support on the impurity, and zero otherwise. We can equivalently express the disorder-averaged OTOC within the medium at position x as $\overline{B^{(M)}(x, t)} = \frac{2}{3} \overline{n_M(x, t)}$, where $B^{(M)}$ is the OTOC within the medium during the backflow protocol. The late-time dynamics of $\overline{n_M(x, t)}$ is characterized by regions of zero and non-zero density separated by diffusively propagating boundaries. The dynamics of these boundaries can be well captured by a hydrodynamic description where the probability distribution of the boundary position evolves according to a Fokker-Planck equation [68]. The OTOC weight at the position of the impurity is given by

$$P(v_d, t) = \overline{n_M(\overline{x_I}, t)} = \frac{1}{2} \left[1 - \operatorname{erf} \left(\frac{v \sqrt{t}}{2\sqrt{D}} \right) \right], \quad (6)$$

where $v = v_d - v_B$ is the difference between the impurity velocity, v_d and operator butterfly velocity v_B , and D is a constant. This satisfies the diffusion-like scaling relation:

$$P(v, t) = F(v\sqrt{t}). \quad (7)$$

The backflow transition satisfies precisely the same scaling relation as that followed by the OTOC weight within the medium at the position of the impurity. This occurs because at late times the impurity matches the free dynamics of the OTOC within the medium